



E-COMMERCE CHECKOUT & CONVERSION OPTIMISATION



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1. Project Summary:

An e-commerce company is experiencing customer drop-offs during the shopping and checkout journey. Customers browse products and add items to their cart, but some do not complete the purchase due to a lengthy checkout process, unexpected costs, payment issues, or limited visibility into order progress.

The proposed analysis focuses on understanding customer behaviour across the shopping funnel, identifying the major drop-off points, and using data to recommend improvements to increase checkout completion and overall conversion.

2. Customer Shopping Journey:

Search Product → View Product → Add to Cart → Checkout
→ Enter Details → Select Payment → Place Order → Order Confirmation

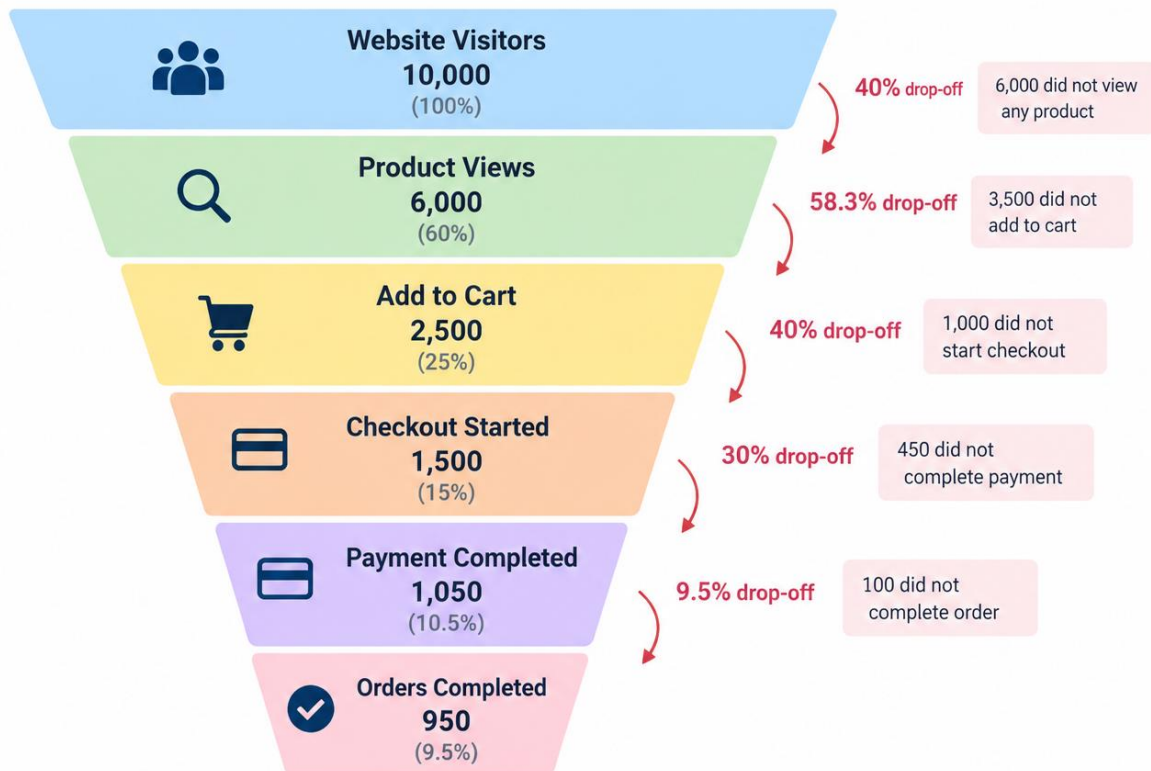
3. Problem & Pain Points:

- High customer drop-off during checkout
- Too many steps to complete a purchase
- Unexpected shipping or other charges
- Payment failures during checkout
- Limited visibility into why customers abandon their carts
- Mobile users experience higher drop-off than desktop users

4. Customer Funnel Analysis:

Customer Funnel Analysis

From website visit to order completion



Key Insight: The largest drop-off occurs between Product Views and Add to Cart (58.3%). This indicates an opportunity to improve product information, pricing, or promotions.

5. SQL Analysis:

Business Question 1: What is the checkout conversion rate?

```
SELECT  
  ROUND(  
    COUNT(CASE WHEN order_status = 'COMPLETED' THEN 1 END) * 100.0  
    / COUNT(*), 2  
  ) AS checkout_conversion_rate  
FROM orders;
```

Illustrative Result: 63.3%

Insight: A significant portion of checkout sessions do not result in completed orders, indicating an opportunity to improve the checkout experience.

Business Question 2: Which payment method has the highest failure rate?

```
SELECT
  payment_method,
  ROUND(
    COUNT(CASE WHEN payment_status = 'FAILED' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS failure_rate
FROM payments
GROUP BY payment_method
ORDER BY failure_rate DESC;
```

Illustrative Insight: One payment method shows a higher failure rate and should be investigated further.

Business Question 3: Which stage has the highest customer drop-off?

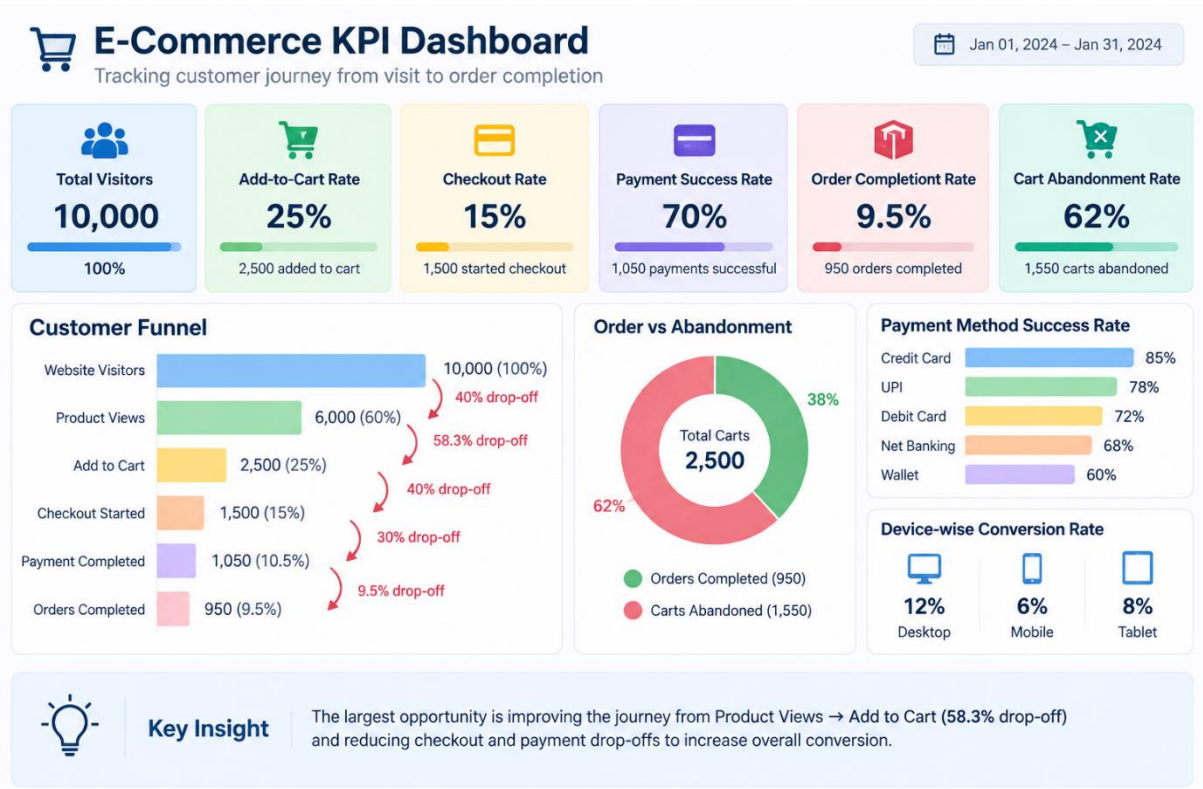
```
SELECT
  stage,
  users,
  previous_stage_users - users AS drop_off_users
FROM funnel_analysis;
```

Illustrative Result: The highest drop-off occurs between **Product View** and **Add to Cart**.

Insight: Investigate product information, pricing, promotions, or customer intent at this stage.

Note: SQL queries, results, and insights are based on illustrative portfolio data.

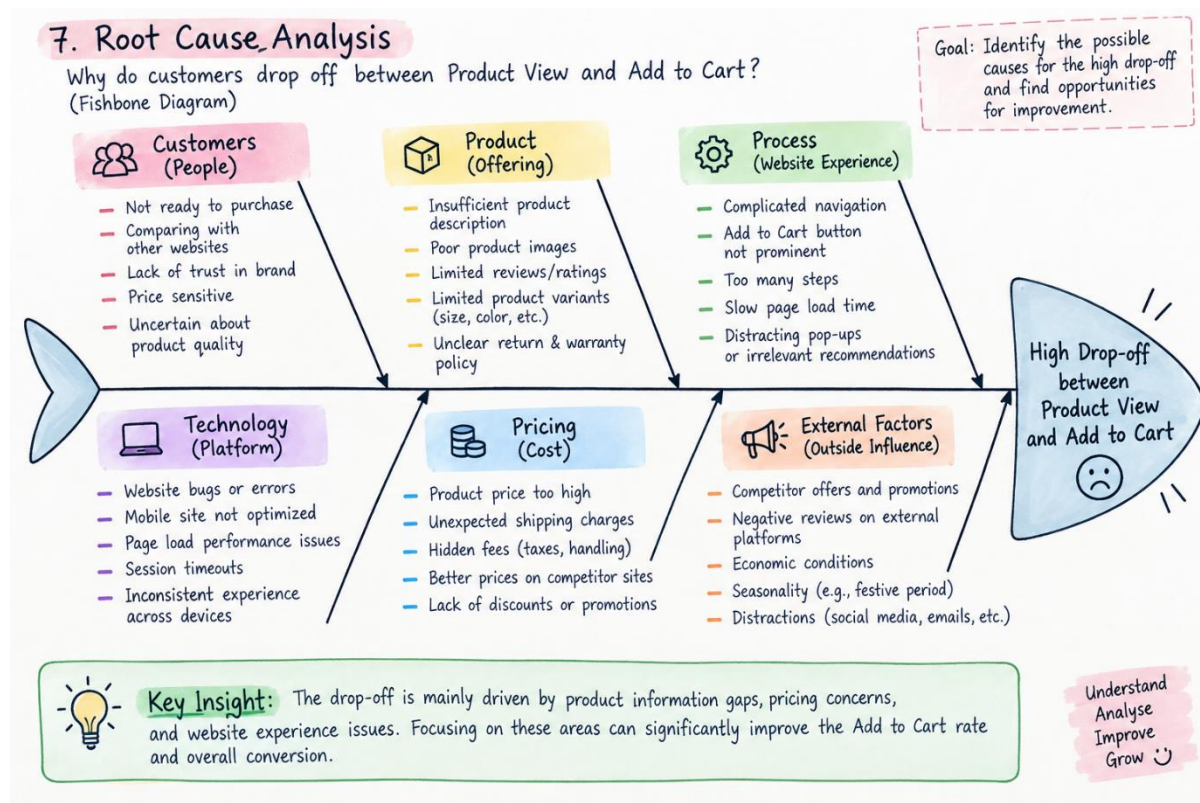
6. KPI dashboard:



7. Root Cause Analysis :

Key Finding

Largest Drop-off: Product View → Add to Cart



Priority Area

Improve the Product Details and Add-to-Cart experience before optimizing later checkout stages.

8. A/B Testing

Test Hypothesis:

Simplifying the product page and making the **Add to Cart** button more visible will increase the Add-to-Cart rate.

	Version A – Current	Version B – Improved
Product Information	Existing layout	Clearer product details
Add to Cart	Less prominent	Prominent button
Reviews	Limited visibility	Visible near product details
Expected Metric	—	Add-to-Cart Rate

Success Metric: Add-to-Cart Conversion Rate

9. Product Recommendations:

Based on the funnel and A/B test findings, the following improvements are recommended:

- Make **Add to Cart** more prominent.
- Display **price, shipping cost and offers** clearly.
- Improve **product images, reviews and key details**.
- Simplify the **mobile shopping experience**.
- Monitor **Add-to-Cart and checkout conversion rates** after changes.

Expected Outcome: Improve product engagement and increase completed purchases.

10. Success Measurement:

After the improvements are implemented, the following metrics will be monitored:

Metric	Purpose
Add-to-Cart Rate	Measure product-page effectiveness
Checkout Conversion Rate	Measure checkout improvement
Payment Success Rate	Track payment performance
Cart Abandonment Rate	Identify remaining drop-offs
Overall Conversion Rate	Measure overall business impact

Success Indicator: Improvement in Add-to-Cart and completed-order conversion compared with the existing baseline.